

# Tender Coconut Classification using Decision Tree and Deep Learning Technique

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**Abstract**— A number of industry sectors such as Food & nutraceuticals, Pharma, Irrigation, Transport and Automobile use coconut and its parts for varied products. As different sectors make use of different type and parts of coconut, an automated separation of coconut would aid in faster processing. Thus the major issue faced by these sectors is the automated sorting of coconut into different categories such as Non-Tender Coconut, Tender Whole coconut, Tender coconut cut with flesh and without flesh. In this work we have trained and tested a Decision Tree and DenseNet model to automatically classify coconut images into various categories. For Decision Tree 93% of accuracy was obtained while the advanced Deep learning model, DenseNet, gave 100% accuracy. This showed that the Deep Learning models such as DenseNet perform better as compared to conventional machine learning algorithms such as Decision Tree.

**Keywords**—Classification, Coconut, Decision Tree, Deep Learning

## I. INTRODUCTION

India is one of the largest producer as well as consumer of coconut in world [1]. Each part of the coconut is useful for different purposes, thus creating opportunities for wide range of industry. Sectors such as Food, Pharmaceuticals, Irrigation, Agriculture, transport and automobile dominate the industries which use this miracle seed [2]. The major issue faced by the foresaid industries is the automated sorting of each seed into different class or types. There is an urgent need for an efficient image based classification of each seed type and synchronization with an automated device for their sorting. Earlier traditional machine learning algorithms such as Support vector machines, Decision trees have been used for classification of fruits and nuts majorly Areca nut. One of the study includes a two class classification system built using red and green color as features. This work comprised of segmentation, masking and classification [3]. A decision tree classifier by considering texture features using GLCM was proposed for classification of arecanut [4]. There are few studies available in literature on classification and grading of different crops using computer vision and machine learning techniques. In another report on categorization of ripening of oil palm fresh fruit bunch using Artificial Neural Network (ANN) was published [5]. Development of machine vision techniques for grading of quality of fruits like Oranges, Peaches and Apples were evaluated [6]. There have been not

many reports or research based on Machine Learning (ML) for classifying coconut. A report that describes classification of coconut based on features such as color and texture is available. It applies technique of machine learning, the Support Vector Machine (SVM) for classification of coconuts into three separate classes, tender coconut (TC), mature coconut (MC) and Copra (CP) [7]. The study is limited to only three features of the coconut and accuracy reported is 99%. The traditional and conventional methods of feature extraction and analysis are dependent on the images available. Hence, the performance of these method depends on the extracted feature values. The current research focuses on the comparison of conventional ML algorithm with Deep learning technique and the latter was observed to give results with 100% accuracy.

In the present work, we trained and tested Decision Tree and DenseNet model from classification of coconut images.

## II. RELATED WORK

The sorting of fruits and nuts via image processing has been one of the most important contributions of ML in Agriculture. Berries and melons and nuts such as areca nut and coconut have been classified using conventional ML techniques for their quality check. There has been a report for identifying coconut by specific features [7] and application of intelligent systems to distinguish between the tender and mature coconut [8]. A report on extraction of coconut images via Gray Level Matrices and K-means for identification of degree of freshness has been published [9]. Significant Markers for determining the level of maturity in coconut based on neural network has been published [10]. The machine learning and deep learning techniques to identify diseases in coconut is published [11]. Prediction of the maturity stages in tender coconut with great accuracy using ANN is reported by N. A. Fadchar and J.C.D. Cruz [12]. Earlier, advanced Intelligent Systems have been implemented which makes use of three important machine learning techniques such as Artificial Neural Network (ANN), Random Forest (RF) and Support Vector Machine (SVM) have been used to classify coconut maturity levels [13]. In one of the publications, classification based on images have been reported for five different types of Nuts viz., Peanut, hazelnut, Pistachios, chestnuts and walnut. The report made use of Convolutional Neural Network

(CNN) algorithms to classify these nuts with the with a dataset of total 2868 images .and the trained model achieved an accuracy of 98% [14]. In yet another report, an application of neural networks and image processing techniques for differentiating healthy and diseased areca nuts has been reported. The distinct classification is based on the defects with diseases or insects of areca nuts that was segmented by a detection line (DL) method. Finally, a back-propagation neural network classifier was applied for detection of quality of areca nuts. The method applied showed the classification of areca nuts to an accuracy of 90.9% [15]. In one of the publications, quality of peanuts was detected and classified based on imaging. Both the color information as well as the texture information correlation were considered as parametric features. The results obtained showed that the method applied resulted in an accuracy ranging from 80.12% to 93.33% for various categories of peanut such as damaged and diseased peanut, for various shape and size peanuts [16]. As per the study by Ruopeng An, *et al.*, AI based neural network model of computer vision was applied to automate nut type classification. The report describes classification of 11 nut types with the dataset containing 2200 images with accuracy ranging from 99.55% to 100% [17].In one of the studies, nuts are categorized using image processing techniques and classified by k-means and k-means++ methods. The paper presented a newer technique called the pixel area and square area method (PASAM), that is apprehended by clustering feature selection of pixel area (PA) and square area (SA) values. The results obtained were with 100% accuracy by PASAM methods [18].

### III. METHODLOGY

In this section we describe our proposed system for classification of coconut.

#### A. Decision Tree

Decision tree is a supervised learning algorithm used for classification and regression. It is used to make predictions based on how a previous set of questions were answered. We have used decision tree to classify the given images into a set of four categories namely, Non-Tender coconut, Tender Whole Coconut, Tender coconut cut with flesh and without flesh.

#### B. DenseNet

DenseNet utilizes connections which are dense between layers and it is a type of convolutional neural network. It helps in solving the problem of vanishing gradient and thereby provides us with an improvement in the declining accuracy.

The architecture of the proposed system for classification of coconut images is shown in Fig. 1. In the proposed architecture we have taken the images of four different categories and we have applied decision tree and deep learning algorithm. The deep learning model which we have used here is the DenseNet model for classification.

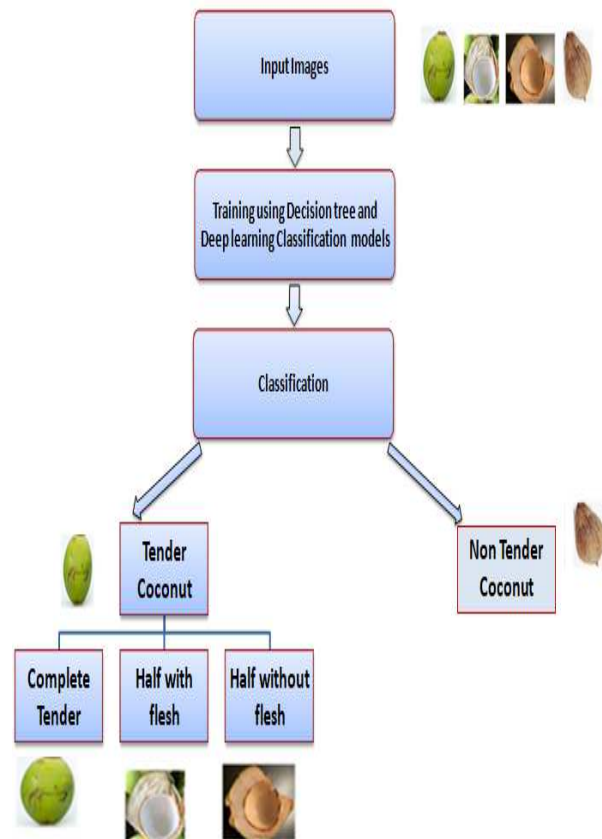


Fig. 1: Proposed System for Classification of Coconut images



a. Non-Tender Coconut



b. Tender coconut with flesh



c. Tender coconut without flesh



d. Tender Whole coconut

Fig. 2. Sample coconut images of four different categories.

#### IV. RESULTS

In this section we describe our dataset and results obtained.

##### A. DataSets

The statistics of the data set used for classification is given in the Table I. These images were taken using mobile phone camera.

TABLE I. COCONUT DATASET STATISTICS

| Label                        | Number of Images |
|------------------------------|------------------|
| Coconut                      | 31               |
| Tender coconut whole         | 220              |
| Tender coconut with flesh    | 74               |
| Tender coconut without flesh | 42               |

##### B. Results

In this section we show the results obtained for DecisionTree algorithm and DenseNet model.

a) Decision Tree Results: Table II shows the Confusion matrix results of Decision Tree algorithm

TABLE II. CONFUSION MATRIX OF DECISION TREE ALGORITHM

| Predicted | Actual       |                        |                           | 3<br>Whole |
|-----------|--------------|------------------------|---------------------------|------------|
|           | 0<br>Coconut | 1<br>Tender-with-flesh | 2<br>Tender-without-flesh |            |
| 0         | 12           | 0                      | 0                         | 0          |
| 1         | 1            | 20                     | 0                         | 1          |
| 2         | 0            | 0                      | 8                         | 0          |
| 3         | 4            | 0                      | 2                         | 59         |

The Precision, Recall, F1 score and accuracy results obtained for Decision Tree is shown in Table III. We obtained an accuracy of 93% for Decision tree algorithm.

TABLE III. CLASSIFICATION REPORT OF DECISION TREE ALGORITHM

| Class            | Precision | Recall | f1-score | Support |
|------------------|-----------|--------|----------|---------|
| 0                | 0.71      | 1.00   | 0.83     | 12      |
| 1                | 1.00      | 0.91   | 0.95     | 22      |
| 2                | 0.80      | 1.00   | 0.89     | 8       |
| 3                | 0.98      | 0.91   | 0.94     | 65      |
| Accuracy         |           |        | 0.93     | 107     |
| Macro average    | 0.87      | 0.95   | 0.90     | 107     |
| Weighted average | 0.94      | 0.93   | 0.93     | 107     |

##### b) DenseNet Model Results

Table. IV shows the Confusion matrix results of DenseNet model.

TABLE IV. CONFUSION MATRIX OF DENSENET MODEL

| Predicted | Actual       |            |                        |                           |
|-----------|--------------|------------|------------------------|---------------------------|
|           | 0<br>Coconut | 1<br>Whole | 2<br>Tender-with-flesh | 3<br>Tender-without-flesh |
| 0         | 5            | 0          | 0                      | 0                         |
| 1         | 0            | 3          | 0                      | 0                         |
| 2         | 0            | 0          | 11                     | 0                         |
| 3         | 0            | 0          | 0                      | 2                         |

The Precision, Recall, F1 score and accuracy results obtained for DenseNet is shown in Table V. We obtained an accuracy of 100% for DenseNet model.

TABLE V: CLASSIFICATION REPORT OF DENSENET MODEL

| GClass           | Precision | Recall | F1-score | Support |
|------------------|-----------|--------|----------|---------|
| 0                | 1.00      | 1.00   | 1.00     | 5       |
| 1                | 1.00      | 1.00   | 1.00     | 3       |
| 2                | 1.00      | 1.00   | 1.00     | 11      |
| 3                | 1.00      | 1.00   | 1.00     | 2       |
| Accuracy         |           |        | 1.00     | 21      |
| Macro average    | 1.00      | 1.00   | 1.00     | 21      |
| Weighted average | 1.00      | 1.00   | 1.00     | 21      |

## V. CONCLUSION

Automated coconut classification will be useful in many industries such as Food, Pharma, Food, Irrigation, transport and automobile for generating by products from parts of coconut. Hence, in this work we have fine tuned Decision tree algorithm and DenseNet model for classification of coconut images as Non-Tender coconut, Tender Whole coconut, Tender Coconut cut with and without flesh. Decision Tree algorithm gave an accuracy of 93% and DenseNet model showed an accuracy of 100%. This shows that the advanced Deep Learning models such as DenseNet perform better in classification of various categories of coconut images. In future, we can integrate the trained DenseNet model for developing a fully automated hardware machine for automatic sorting of coconuts into various categories.

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